Fixing a MIMIC-IV benchmark: a leakage-free, landmark-based model for one-year mortality in ICU patients with diabetic neuropathy

BMI5101 Group Project (P16)

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Abstract

Background. Huang et al. (2025) built machine-learning models for mortality in intensive care unit (ICU) patients with diabetic neuropathy (DN) from MIMIC-IV. After LASSO selection of 12 of 55 candidate predictors, random forest (RF) outperformed XGBoost, support vector machine and logistic regression (LR) (validation AUC 0.780). However, inclusion used information from after the first ICU day, the outcome was not consistently defined, and models were chosen on one 263-patient validation set with selection and tuning outside cross-validation.

Methods. Keeping the benchmark's framework, we used MIMIC-IV v3.1 to build a landmark model predicting at 24 h after ICU admission from information then available. We retained patients without a white-cell differential, separated chronic from acute comorbidity codes and defined death within 365 days. Models were evaluated by 5×10 repeated nested cross-validation with imputation, LASSO and tuning inside each fold, and chosen by a pre-specified rule; the selected model was frozen and evaluated once on a held-out set. A reference cohort with the benchmark's criteria tested the original pipeline. As exploratory extensions, first-day trajectory features, prior-diagnosis embeddings and stacking were compared on the development set.

Results. In the reference cohort, the benchmark's RF reproduced its training AUC of 0.999 but reached only 0.724–0.767 under nested cross-validation, against 0.791 for the best model. In the landmark cohort (2,557 patients, 26.6% mortality), six configurations were equivalent and L2-penalised LR was selected (nested-CV AUROC 0.801). Its nested-CV AUPRC was 0.594 (prevalence 0.266). On the 512-patient held-out set its AUROC was 0.750 (95% CI 0.702–0.797), its AUPRC 0.533 (0.459–0.611) and its calibration slope 0.82. Red cell distribution width, Charlson comorbidity index and age were the leading predictors. Stacking raised AUROC by 0.005 over LR (95% CI -0.001 to 0.011), and trajectory features and diagnosis embeddings gave no gain.

Conclusions. Under leakage-free evaluation, LR matches tree ensembles and the benchmark's RF advantage is not supported. The landmark model can be applied at a defined time but needs local recalibration before use.

Keywords: diabetic neuropathy; ICU; mortality prediction; MIMIC-IV; landmark model; nested cross-validation; LASSO; calibration

1 Introduction

1.1 Background

Diabetic neuropathy is the most common chronic complication of diabetes and is associated with foot ulceration, amputation, cardiovascular autonomic dysfunction and increased mortality (Feldman et al., 2019; Pop-Busui et al., 2017). Patients with DN who are admitted to the ICU typically carry a heavy burden of long-standing diabetes and comorbid cardiovascular and renal disease. Identifying those at highest risk of death in the year after ICU admission could support goals-of-care discussions and targeted follow-up.

1.2 The benchmark study

Huang et al. (2025) used the MIMIC-IV database (Johnson et al., 2023) to predict death in ICU patients with DN. Their framework had five steps: an ICD-defined DN cohort; 55 candidate predictors from the first ICU day (demographics, comorbidities, laboratory values, vital signs, severity scores and inflammatory ratios); LASSO selection of 12 predictors (Tibshirani, 1996); RF, XGBoost, SVM and LR models; and SHAP interpretation of the best model (Lundberg & Lee, 2017), supported by decision-curve and clinical-impact analyses. RF was reported as the best model (validation AUC 0.780), and RDW, neutrophil count, the Charlson comorbidity index (CCI), chloride and prothrombin time were the most important predictors. The framework is reasonable and widely used, but several design choices limit what its results can show (Collins et al., 2015).

1.3 Issues and our modifications

Table 1 lists seven issues in the benchmark, what each implies, and how we changed the analysis. The rest of the paper follows this table: Section 2 implements each modification, Section 3 gives the evidence, and Section 4 summarises the effect of each change.

Table 1: Issues in the benchmark and our modifications.

#	Issue in the benchmark	Consequence	Our modification	Sections
I1	Inclusion required a hospital stay longer than 48 h.	Length of stay is unknown on the first ICU day; early deaths and early discharges are excluded, and the model has no defined time of use.	Fixed prediction time at 24 h after ICU admission (landmark); all adults alive and in the ICU at that time are included.	2.2, 3.1
I2	All eight complete-blood-count items were required.	About one third of eligible patients are excluded, almost all because no white-cell differential was performed.	These patients are kept; a missing-differential indicator is added and values are imputed within folds.	2.3, 3.1, 3.6
I3	Comorbidities were taken from all codes of the index admission.	Untimed codes for acute events of the current stay are counted as pre-existing disease.	Comorbidities from prior admissions plus chronic index-admission codes; CCI without its age component.	2.3, 3.6

#	Issue in the benchmark	Consequence	Our modification	Sections
I4	The outcome was described inconsistently (in-hospital death in the abstract, a death within one year after discharge in a SHAP example) and was not defined in the Methods; white-cell differential units appear mixed; the hypertension definition was not given.	Mortality cannot be interpreted; differential-based predictors may reflect test format; a narrow hypertension definition creates a spurious protective effect.	Death within 365 days of ICU admission; absolute cell counts; full Quan–Elixhauser hypertension definition.	2.3, 3.2
I5	Models were compared on one 263-patient validation set; LASSO and tuning were run on the whole training set outside cross-validation.	Model differences are smaller than the uncertainty of a single small validation set, and performance estimates are optimistic (Cawley & Talbot, 2010; Varma & Simon, 2006).	Repeated nested cross-validation with every data-dependent step inside the folds; corrected resampled t-test; pre-specified selection rule; frozen model evaluated once on a held-out set.	2.5, 3.3, 3.4
I6	RF was chosen despite a training AUC of 0.999 against a validation AUC of 0.780; tuning grids were not reported.	The selected model is likely overfitted, and comparator models may not have been tuned comparably.	Every model family tuned on stated grids inside the folds and compared side by side with the benchmark's parameters.	2.4, 3.3
I7	Performance was reported as AUC, threshold metrics, decision curves and clinical impact curves on the validation set, without calibration or Brier score; the threshold metrics appear consistent only with survival as the positive class.	Whether predicted risks are accurate is unknown, and the classification results do not describe the detection of deaths.	Calibration intercept and slope, Brier and scaled Brier scores and AUPRC added; death as the positive class with a threshold from out-of-fold predictions; decision curves on cross-validated and frozen held-out predictions.	2.6, 3.5

1.4 Contributions

This study makes four contributions:

1. **A leakage-free evaluation of the benchmark framework.** Repeated nested cross-validation, a corrected test and a pre-specified selection rule are applied to the benchmark's own model configurations and to tuned alternatives.
2. **A deployable landmark model.** The model predicts at a defined time from information available then, retains patients without a white-cell differential and uses timed comorbidity information; it was frozen before a single confirmatory held-out evaluation.
3. **An empirical check of the benchmark's issues.** For each issue in Table 1 we report how much the modification changed the cohort, the predictors or the performance, including the modifications that made little difference.
4. **An exploratory test of proposed extensions.** First-day trajectory features, prior-diagnosis representations and a stacking ensemble are compared with matched controls on the same folds.

2 Methods

2.1 Data source and the framework we build on

We used MIMIC-IV version 3.1 (Johnson et al., 2023), which contains de-identified records of patients admitted to the Beth Israel Deaconess Medical Center (Boston, USA) between 2008 and 2022; the benchmark used an earlier v2.x release covering 2008–2019. Severity scores, the comorbidity index and first-day laboratory and vital-sign summaries were computed with the official MIT-LCP `mimic-code` concepts. Analyses were run in Python 3.11 with DuckDB 1.5.6, scikit-learn 1.9, xgboost 3.4 and shap 0.51. In line with the PhysioNet data use agreement, only aggregate results are reported.

We kept the benchmark's framework: the ICD-defined DN population, the 55 candidate predictors from the first ICU day, LASSO selection at λ.1se, the four model families (RF, XGBoost, SVM, LR), the benchmark's reported hyperparameters as one model configuration, SHAP interpretation and decision-curve analysis. The changes are those listed in Table 1 and are described in Sections 2.2–2.6.

2.2 Landmark cohort (I1)

The study population consisted of patients with at least one hospital admission carrying an ICD-9 or ICD-10 code for DN and an ICU stay. For each patient, the **index admission** was the first admission with both a DN code and an ICU stay, and the index ICU stay was the first ICU stay of that admission. The benchmark's wording ("first ICU admission at first hospitalization") admits several readings; readings that require the very first admission or the first-ever ICU stay to qualify exclude eligible patients because of their care history, and are summarised in Supplementary Table S1.

Prediction was made at a fixed **landmark of 24 h after ICU admission** (van Houwelingen, 2007). Patients aged 18 years or older who were still in the ICU and alive at that time were included. Patients whose first ICU stay ended within 24 h were excluded, and a later ICU stay of the same admission was not used in its place. No length-of-stay or blood-count criterion was applied, because neither is known, or clinically required, at the time of prediction. The landmark cohort was split once into a stratified 80% training set and a 20% held-out set.

For comparison, a **reference cohort** applied the benchmark's criteria (hospital stay longer than 48 h, age over 18 years and all eight complete-blood-count items) to the same index admissions; it is analysed in Section 3.3 and the Supplement.

2.3 Outcome and predictors (I2, I3, I4)

Outcome (I4). The outcome was all-cause death on a date between the day of ICU admission and 365 days later, using the in-hospital time of death when available and the date of death from state records otherwise, compared at date granularity. MIMIC-IV records out-of-hospital deaths up to one year after a patient's last discharge, so every death within the 365-day window is observed. Cause of death is not available.

Candidate predictors. We extracted the benchmark's 55 candidate predictors: age and sex; six co-morbidities (myocardial infarction, congestive heart failure, chronic pulmonary disease, renal disease,

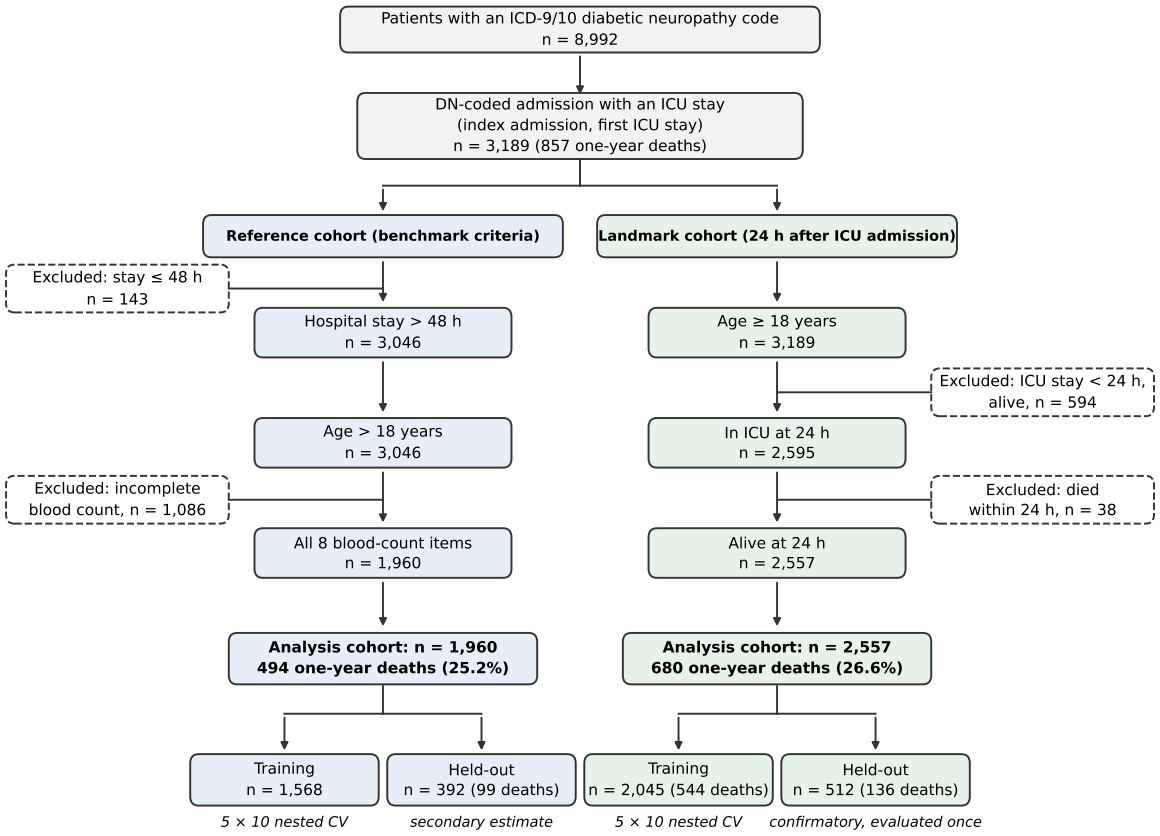


Figure 1: Cohort flow. Both cohorts start from the same index admissions. The reference cohort applies the benchmark's inclusion criteria; the landmark cohort includes all adults alive and in the ICU 24 h after admission.

hypertension, obesity); 31 laboratory summaries (glucose, anion gap, bicarbonate, creatinine, urea nitrogen, haemoglobin, platelets, white and red cells, RDW, absolute lymphocytes, neutrophils and monocytes, calcium, chloride, sodium, potassium, PTT, PT, INR); four vital signs (heart rate, mean blood pressure, respiratory rate, temperature); six severity scores (SAPS-II, APS-III, LODS, OASIS, GCS, CCI); and six inflammatory ratios (NLR, PLR, LMR, PNR, PWR, SII). Laboratory values were taken from 6 h before to 24 h after ICU admission and vital signs from the first 24 h, so all predictors are available at the landmark. Every laboratory item identifier was checked against the MIMIC-IV dictionaries.

Units and definitions (I4). White-cell differentials were taken as absolute counts (10^9/L) from the `mimic-code` blood-differential concept, so that no patient's value mixes absolute counts with percentages. Hypertension followed the Quan–Elixhauser definition including complicated hypertension (ICD-9 401–405; ICD-10 I10–I13, I15, I16) (Quan et al., 2005). Under ICD rules, hypertension with chronic kidney disease or heart failure is coded under the complicated codes alone, so an uncomplicated-only definition labels these patients as normotensive.

Comorbidity timing (I3). Diagnosis codes in MIMIC-IV carry no timestamps, so codes of the index admission may describe acute events of the current stay. In the landmark cohort, comorbidities were derived from all codes of prior admissions plus the index-admission codes that remained after removing acute-event codes (for example acute myocardial infarction, acute heart failure, acute stroke or transient ischaemic attack, aortic dissection, acute peptic ulcer and shock liver). The CCI was computed without its age component, because age enters the model separately, and the chronic-disease item of SAPS-II was recomputed over prior and index admissions.

White-cell differential (I2). Instead of excluding patients without a differential, we added one binary candidate predictor, `wbc_diff_missing`, equal to 1 when all three absolute cell counts were missing. The eight differential-derived variables (three cell counts and NLR, PLR, LMR, PNR, SII) were imputed within folds like every other variable. The landmark analysis therefore had 56 candidate predictors.

2.4 Models (I6)

Preprocessing. Within each training set, variables with more than 20% missing values were dropped (the eight differential-derived variables were exempt in the landmark cohort). Missing values were then imputed with the training mean for continuous variables and the training mode for binary variables and ordinal scores, and each variable was standardised to zero mean and unit variance using training statistics. LASSO, LR and SVM used standardised values; RF and XGBoost used imputed values on the original scale.

Logistic regression (final model). For patient i with predictor vector $\mathbf{x}_i$, the model is

$$p_i = \sigma(\beta_0 + \mathbf{x}_i^\top \boldsymbol{\beta}), \qquad \sigma(z) = \frac{1}{1+e^{-z}}.$$

LASSO selection. Variables were selected by L1-penalised LR (Tibshirani, 1996), which solves

$$\min_{\beta_0,\boldsymbol{\beta}} -\frac{1}{n}\sum_{i=1}^{n}\left[y_i \log p_i + (1-y_i)\log(1-p_i)\right] + \lambda\|\boldsymbol{\beta}\|_1,$$

with the intercept unpenalised. We used scikit-learn's `liblinear` solver, which minimises $\|\boldsymbol{\beta}\|_1 + C\sum_i \ell_i$, where ℓ_i is the log-loss of patient i. Dividing by nC shows that the two problems are identical when $C = 1/(n\lambda)$, and this conversion was used for every λ (implementation details in Supplementary S9).

The solver (newGLMNET) is a coordinate-descent method of the glmnet type (Friedman et al., 2010; Yuan et al., 2012). At each outer iteration it replaces the log-likelihood by a weighted least-squares approximation around the current estimate, with working responses z_i and weights $w_i = \tilde{p}_i(1-\tilde{p}_i)$, and then updates one coefficient at a time. The update for coefficient j is

$$\beta_j \leftarrow \frac{S\big(\frac{1}{n}\sum_i w_i x_{ij} r_{ij},\ \lambda\big)}{\frac{1}{n}\sum_i w_i x_{ij}^2}, \qquad S(z,\lambda) = \operatorname{sign}(z)\,(|z|-\lambda)_+,$$

where r_{ij} is the partial residual of z_i that excludes predictor j. The soft-threshold operator S returns exactly zero whenever $|z| \le \lambda$, that is, whenever the weighted association between predictor j and the partial residual is too weak to offset the penalty. Coefficients are therefore set exactly to zero rather than merely shrunk, and the predictors with non-zero coefficients form the selected set.

Choice of λ. Over a glmnet-style grid of λ values (Supplementary S9), stratified 10-fold cross-validation gave the mean held-out binomial deviance $\mathrm{CV}(\lambda)$ and its standard error $\mathrm{SE}(\lambda)$, the standard deviation across folds divided by $\sqrt{10}$. With $\lambda_{\min} = \arg\min_\lambda \mathrm{CV}(\lambda)$, the **$\lambda$.1se rule** takes the largest λ with $\mathrm{CV}(\lambda) \le \mathrm{CV}(\lambda_{\min}) + \mathrm{SE}(\lambda_{\min})$, the sparsest model whose deviance is within one standard error of the minimum. The predictors with non-zero coefficients at λ.1se, refitted on the whole training data, were retained.

L2-penalised and unpenalised LR. On the selected predictors, the L2-penalised LR solves

$$\min_{\beta_0,\boldsymbol{\beta}} \frac{1}{n}\sum_{i=1}^{n} \ell_i + \frac{\lambda_2}{2}\|\boldsymbol{\beta}\|_2^2.$$

scikit-learn minimises $\frac{1}{2}\|\boldsymbol{\beta}\|_2^2 + C\sum_i \ell_i$ (L-BFGS solver, intercept unpenalised), which is the same problem with $\lambda_2 = 1/(nC)$. C was tuned over $\{0.001, 0.01, 0.1, 1, 10, 100, \infty\}$, where $C = \infty$ removes the penalty. The **unpenalised LR**, which corresponds to the benchmark's LR fitted with R `glm`, was also fitted as a separate configuration.

Random forest. An RF averages B classification trees, each grown on a bootstrap sample with a random subset of predictors considered at each split (Breiman, 2001):

$$\hat{p}(\mathbf{x}) = \frac{1}{B}\sum_{b=1}^{B} T_b(\mathbf{x}),$$

where $T_b(\mathbf{x})$ is the proportion of deaths in the leaf of tree b that contains $\mathbf{x}$. The tuned RF used $B = 500$ and a grid over the number of predictors per split, the minimum leaf size and the maximum depth.

XGBoost. XGBoost is an additive tree model on the log-odds scale, $\operatorname{logit}\hat{p}(\mathbf{x}) = \sum_{k=1}^{K} f_k(\mathbf{x})$, fitted by gradient boosting with shrinkage (learning rate η) to minimise $\sum_i \ell_i + \sum_k \Omega(f_k)$ with

$$\Omega(f) = \gamma T + \tfrac{1}{2}\lambda\|\mathbf{w}\|^2,$$

where T is the number of leaves and $\mathbf{w}$ the leaf weights (Chen & Guestrin, 2016). γ and λ were kept at $\gamma = 0$ and the library default $\lambda = 1$; the number of trees, depth, learning rate, column and row subsampling and minimum child weight were tuned.

Support vector machine. The SVM solves the soft-margin problem $\min_{\mathbf{w},b,\boldsymbol{\xi}} \tfrac{1}{2}\|\mathbf{w}\|^2 + C\sum_i \xi_i$ subject to $\tilde{y}_i(\mathbf{w}^\top \phi(\mathbf{x}_i) + b) \geq 1 - \xi_i$ and $\xi_i \geq 0$, with $\tilde{y}_i \in \{-1, +1\}$ and the radial basis function kernel $K(\mathbf{x}, \mathbf{x}') = \exp(-\gamma\|\mathbf{x} - \mathbf{x}'\|^2)$ (Cortes & Vapnik, 1995). The SVM outputs a decision value $f(\mathbf{x})$; probabilities were obtained by Platt scaling, $\hat{p} = 1/(1 + \exp(Af(\mathbf{x}) + B))$, with A and B fitted by internal cross-validation (`probability=True`) (Platt, 1999). During tuning, AUROC was computed from the decision values, and the selected configuration was then refitted with Platt scaling. C and γ were tuned.

Two configurations per family. Each family was run (a) with the hyperparameters reported by the benchmark and (b) tuned by grid search inside the training data (stratified 5-fold cross-validation, AUROC as the criterion). Together with the unpenalised LR this gives nine configurations. RF was run in two benchmark versions, one with the literal parameters and one with R `randomForest` defaults (nine fully grown trees), because the benchmark lists R and Python parameter names together and the two readings imply very different trees. All grids are listed in Supplementary S9.

2.5 Evaluation framework (I5)

Repeated nested cross-validation (main estimate). Each training set was split into 10 stratified outer folds, and the split was repeated 5 times with different seeds, giving $J = 50$ outer folds. Inside every outer training fold, the 20% missingness rule, imputation, standardisation, LASSO (inner stratified 10-fold cross-validation, λ.1se) and hyperparameter tuning (inner stratified 5-fold cross-validation) were run in turn; the model was then refitted on the whole outer training fold and evaluated on the untouched outer validation fold (Figure 2). This estimates the performance of the whole modelling procedure, including variable selection and tuning, rather than of one fitted model (Cawley & Talbot, 2010; Varma & Simon, 2006). AUROC is reported as the mean $\pm$ SD over the 50 outer folds. In each repeat the 10 outer validation folds were pooled into out-of-fold (OOF) predictions covering the whole training set; AUPRC, Brier score, calibration, decision curves and thresholds were computed from these and averaged over the 5 repeats.

Model comparison. Two models evaluated on the same folds were compared with the Nadeau–Bengio corrected resampled t-test (Nadeau & Bengio, 2003). With d_j the AUROC difference in outer fold j, $\bar{d}$ its mean and $\hat{\sigma}_d^2$ the sample variance of the J differences,

$$t = \frac{\bar{d}}{\sqrt{\left(\frac{1}{J} + \frac{n_{\text{test}}}{n_{\text{train}}}\right)\hat{\sigma}_d^2}},$$

referred to a t distribution with $J - 1 = 49$ degrees of freedom, where $n_{\text{test}}/n_{\text{train}}$ is the ratio of validation-fold to training-fold size. The term $n_{\text{test}}/n_{\text{train}}$ inflates the variance to account for the overlap between

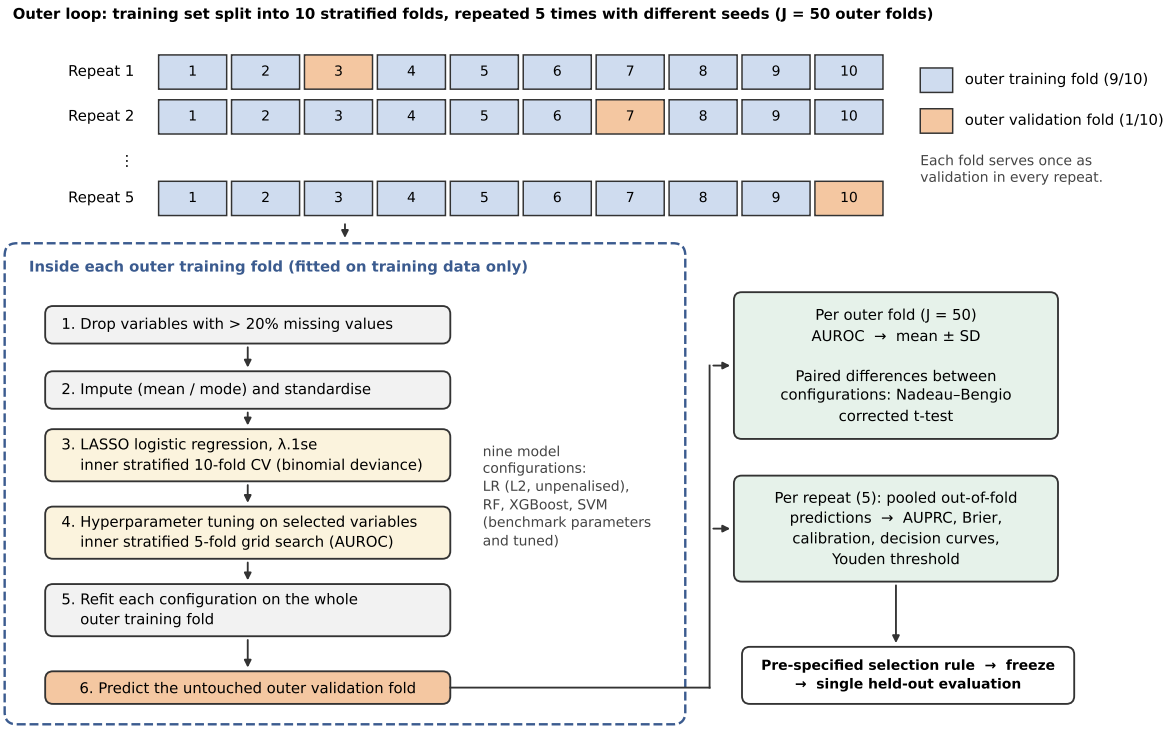


Figure 2: Repeated nested cross-validation. Every data-dependent step, including variable selection and tuning, is fitted inside each outer training fold and evaluated on the untouched outer validation fold.

training sets, which the naive paired t-test (variance $\hat{\sigma}_{\bar{d}}^2/J$) ignores. The 95% confidence interval for the difference is $\bar{d} \pm t_{0.975,J-1}$ times the corrected standard error. In a simulation with 200 null datasets and the same 10-fold, 5-repeat design, the corrected test rejected the null hypothesis in 4.0% of datasets and the uncorrected paired t-test in 38.0%.

Pre-specified model-selection rule. (1) The configuration with the highest mean outer AUROC is the reference M*. (2) A configuration is equivalent to M* if its AUROC difference is at most 0.01, or at most 0.02 with a 95% confidence interval that includes zero. (3) If the equivalent set contains an LR, an LR is chosen, for parsimony and interpretability. (4) If both LRs are equivalent, the one whose pooled-OOF calibration slope is closer to 1 is chosen, measured by the mean over repeats of |ln slope| (the unpenalised LR on a tie). (5) The held-out set plays no part. The rule was written before any landmark analysis was run; for the reference cohort it was written after exploratory analyses and is therefore not blind.

Freezing and the held-out evaluation. After selection, the final landmark model was fitted on the whole training set with the same pipeline (LASSO with inner 10-fold cross-validation, then C chosen by inner 5-fold cross-validation). Its specification, the classification threshold and identifiers of the training data and code were recorded under version control before the held-out set was read. The held-out evaluation was run exactly once and was not repeated or followed by changes to the model. A tuned RF, frozen in the same way, served as a comparator. In the reference cohort the held-out set had been used in earlier exploratory model comparisons, so its results are reported only as a secondary estimate.

Held-out uncertainty. On held-out sets, the AUROC 95% confidence interval used the DeLong variance (DeLong et al., 1988), computed with the fast algorithm of Sun and Xu (2014), and the two frozen models

were compared with the paired DeLong test. AUPRC confidence intervals came from a stratified bootstrap with 2,000 resamples (percentile method).

2.6 Metrics (I7)

Death was the positive class throughout.

Discrimination. AUROC, and AUPRC computed as average precision. The AUPRC of an uninformative model equals the prevalence $\bar{y}$, which is reported as its baseline.

Overall accuracy. The Brier score and the scaled Brier score (Steyerberg et al., 2010),

$$\text{Brier} = \frac{1}{n}\sum_{i=1}^{n}(\hat{p}_i - y_i)^2, \qquad \text{scaled Brier} = 1 - \frac{\text{Brier}}{\bar{y}(1-\bar{y})},$$

where $\bar{y}(1-\bar{y})$ is the Brier score of a model that predicts the prevalence for everyone, computed on the same data.

Calibration (Van Calster et al., 2019). A logistic recalibration model

$$\operatorname{logit} P(y=1) = a + b \operatorname{logit} \hat{p}$$

was fitted to the predictions. The calibration slope is b from fitting a and b together; the calibration intercept (calibration-in-the-large) is a from refitting with b fixed at 1, that is, with $\operatorname{logit}\hat{p}$ as an offset. A well-calibrated model has $a = 0$ and $b = 1$; $b < 1$ means predictions that are too extreme and $b > 1$ predictions that are too close to the mean.

Clinical utility. Decision-curve analysis (Vickers & Elkin, 2006) compares the net benefit of acting on the model at threshold probability p_t,

$$\text{NB}(p_t) = \frac{TP}{n} - \frac{FP}{n} \cdot \frac{p_t}{1-p_t},$$

with treating everyone, $\text{NB}_{\text{all}}(p_t) = \bar{y} - (1-\bar{y})\,\frac{p_t}{1-p_t}$, and treating no one (NB = 0). We evaluated the 56 thresholds from 0.05 to 0.60 in steps of 0.01 and counted the thresholds at which the model's net benefit exceeded both alternatives.

Classification. The threshold maximised Youden's index (Youden, 1950), $J = \text{Se} + \text{Sp} - 1$, on the pooled OOF predictions of each repeat; the median over the 5 repeats was carried unchanged to the held-out set. We report sensitivity, specificity, precision, F1 and the share of patients flagged as high risk.

2.7 Interpretation and sensitivity analyses

SHAP. For an LR on standardised predictors, the SHAP value of predictor j (under feature independence) is

$$\phi_j(\mathbf{x}) = \beta_j\,(x_j - \mathrm{E}[x_j]),$$

on the log-odds scale, with the expectation taken over the training data used as background (Lundberg & Lee, 2017). Because standardisation was fitted on the same training data, $\mathrm{E}[x_j] = 0$ and $\phi_j(\mathbf{x}) = \beta_j x_j$. Global importance is the mean $|\phi_j|$ over the held-out set. Tree models were explained with TreeExplainer.

Odds ratios. For inference, the selected predictors were refitted by unpenalised LR on the training set. Because predictors are standardised, $\exp(\beta_j)$ is the odds ratio per standard deviation, with Wald 95% confidence interval $\exp(\beta_j \pm 1.96\,\mathrm{SE}_j)$; for binary predictors the coefficient was rescaled by the predictor's standard deviation to give the odds ratio for yes versus no. Collinearity was checked with variance inflation factors (VIF).

Sensitivity analyses. All were run with the full nested cross-validation on the same outer splits as the main analysis, and compared with it by the Nadeau–Bengio test: (i) reference cohort, comorbidities and CCI recomputed from prior plus chronic index-admission codes (I3); (ii) reference cohort, exclusion of patients admitted in 2020–2022, who are not covered by the benchmark's data release (new single 80/20 split); (iii) landmark cohort, comorbidities from prior admissions only (*prior-only*, the most conservative deployment scenario; I3); and (iv) landmark cohort, removal of all white-cell differential variables (*no-WBC-differential*; I2).

2.8 Exploratory extensions

The project brief proposed temporal features, comorbidity representations and a stacking ensemble. These were evaluated after the main analysis had been completed, on the landmark development set only (2,045 patients, 544 deaths); the held-out set was not used. Three additions were made to the 56 static candidates. (i) *First-day trajectories*: five statistics (first-to-last change, hourly slope, standard deviation, late-minus-early mean and number of measurement times) for each of 25 laboratory and vital-sign series recorded 0–24 h after ICU admission, giving 125 candidates. (ii) *Prior-diagnosis burden*: four variables describing earlier admissions and their ICD codes. (iii) *Diagnosis embedding*: each patient's ICD codes from admissions that ended before the index admission were converted to a TF-IDF document and compressed to 32 dimensions by truncated singular value decomposition, both fitted within training folds. Stacking combined LR, RF and XGBoost base learners through an L2 LR meta-model trained on their internal 3-fold out-of-fold predictions.

To keep the computation feasible, all extension models and their static controls used one fixed-parameter recipe (LR with $C = 1$; RF with 200 trees of maximum depth 10; XGBoost with 100 trees of depth 2; LASSO with 5 internal folds and 30 values of λ) on the same 5×10 outer splits as the main analysis, with missingness screening, imputation, LASSO and the embedding refitted within each training fold. Each extension was compared with its matched control on the same folds by the corrected test, with Holm adjustment over 15 comparisons per metric. Because this recipe omits inner tuning, the extensions are compared only with their own static controls, not with Table 4. The extension experiments were run on a separate machine; rebuilding their input features from our own data reproduced the same patients, outcomes and missingness pattern, with numerical differences only at floating-point precision (maximum absolute difference 2.8×10^{-14}).

3 Results

3.1 Cohort: who the benchmark's criteria exclude (I1, I2)

This section shows which patients the length-of-stay and blood-count criteria remove (I1, I2).

Of 8,992 patients with a DN code, 3,189 had a DN-coded admission with an ICU stay (Figure 1). The landmark cohort comprised 2,557 patients with 680 deaths within one year (26.6%): 594 patients whose first ICU stay ended within 24 h and 38 who died within 24 h were excluded. The reference cohort, built with the benchmark's criteria, comprised 1,960 patients with 494 deaths (25.2%).

The retention at each benchmark step closely matched the benchmark (hospital stay > 48 h, 95.5% vs 95.4%; complete blood count, 64.3% vs 64.2%), and the larger cohort (1,960 vs 1,313 in the benchmark) probably reflects the extended data release (12.6% of the reference cohort was admitted in 2020–2022). Exclusion at the blood-count step was almost entirely due to a missing white-cell differential (35.6% of candidates); the core blood count was missing in only 1%. The landmark cohort keeps these patients: 873 (34.1%) had no differential, and their one-year mortality was 28.6% (250/873), against 25.5% (430/1,684) for patients with a differential (χ^2 p = 0.09). In total, 1,452 landmark patients (56.8%) had a prior admission from which comorbidities could be taken.

Non-survivors were older and had higher RDW, CCI and severity scores and lower chloride (Table 2).

Table 2: Selected baseline characteristics of the landmark cohort by one-year outcome (median [IQR] or n [%]; observed values before imputation). Full tables for both cohorts are in Supplementary S3.

Variable	Overall (n = 2,557)	Survivors (n = 1,877)	Non-survivors (n = 680)	p
Age, years	67 (58–75)	65 (57–73)	71 (63–79)	<0.001
Male sex	1,520 (59.44%)	1,097 (58.44%)	423 (62.21%)	0.087
Congestive heart failure	1,178 (46.07%)	775 (41.29%)	403 (59.26%)	<0.001
Renal disease	1,299 (50.80%)	877 (46.72%)	422 (62.06%)	<0.001
Obesity (coded)	859 (33.59%)	664 (35.38%)	195 (28.68%)	0.002
CCI (without age)	5 (4–7)	5 (3–6)	6 (5–8)	<0.001
RDW, %	14.65 (13.60–16.20)	14.30 (13.40–15.65)	15.81 (14.49–17.63)	<0.001
Chloride (max), mmol/L	105 (100–109)	106 (101–109)	103 (98–108)	<0.001
Anion gap (min), mmol/L	13 (10–15)	12 (10–14)	14 (11–17)	<0.001
Urea nitrogen (min), mg/dL	23 (15–39)	21 (14–34)	32 (20–53)	<0.001
Lymphocytes, 10^9/L †	1.27 (0.79–1.95)	1.39 (0.87–2.08)	0.95 (0.60–1.47)	<0.001
Respiratory rate, /min	18.28 (16.43–20.62)	18.08 (16.32–20.20)	19.12 (16.99–21.75)	<0.001
SAPS-II	36 (29–45)	34 (27–42)	42 (35–52.25)	<0.001
APS-III	45 (35–57)	42 (34–53)	54 (43–67)	<0.001
OASIS	30 (25–36)	30 (24–35)	33.5 (27–40)	<0.001
No white-cell differential	873 (34.14%)	623 (33.19%)	250 (36.76%)	0.092
Prior admission	1,452 (56.79%)	1,028 (54.77%)	424 (62.35%)	<0.001

† 34.1% missing (no differential performed).

3.2 Definition errors in the benchmark (I4)

This section gives the evidence for the three definition problems addressed by I4. The comparisons use the reference cohort, which follows the benchmark's criteria.

Mortality. One-year mortality from ICU admission was 25.2% in the reference cohort, against 38.2% in the benchmark. In-hospital mortality was only 7.1%, so the benchmark's figure cannot be in-hospital mortality, although its abstract names in-hospital death. Counting every recorded date of death regardless of timing gave 42.8%, the same order as the benchmark, and 49% of patients with an out-of-hospital date of death died more than 365 days after ICU admission. The benchmark may therefore have used an outcome without a fixed follow-up window; because it does not state its definition, this cannot be confirmed.

White-cell differential units. The benchmark reported lymphocytes of 3.69 (1.27–118.32), neutrophils of 23.90 (8.63–831.30) and monocytes of 2.00 (0.60–43.64) $\times$ 10^9/L, which are physiologically implausible. With absolute counts we obtained 1.24 (0.77–1.91), 8.59 (5.99–12.57) and 0.55 (0.35–0.84). The pattern of the derived ratios indicates mixing of scales across patients. Ratios that are unaffected when a patient's three cell counts share one scale agreed with the benchmark (NLR 6.82 vs 6.80; LMR 2.31 vs 2.42; SII 1,336 vs 1,342), whereas ratios that combine platelets with a differential count did not (PLR 159 vs 36; PNR 22 vs 6.6). When we deliberately mixed all differential items, PLR and PNR fell to 24.00 and 3.67, moving towards the benchmark's values. With correct units, neutrophils were selected by LASSO only at λ.min, not at λ.1se, and were not among the five most important predictors, whereas they ranked second in the benchmark.

Hypertension. Hypertension prevalence was 84.6% under the full Quan–Elixhauser definition, against 37.3% in the benchmark; counting only uncomplicated hypertension codes gave 35.4%, close to the benchmark. The narrow definition reverses the apparent effect. Under the full definition, hypertension was unrelated to one-year death (unadjusted OR 1.07, 95% CI 0.80–1.42). Under the uncomplicated definition it appeared protective (OR 0.49, 0.39–0.62), because the 964 patients who had only complicated codes, and were therefore coded as normotensive, had a one-year mortality of 31.3% (OR 1.91, 1.55–2.35). LASSO did not select hypertension under either definition, and the selected variables were identical, so the models were unaffected.

3.3 The benchmark pipeline under leakage-free evaluation (I5, I6)

This section applies the improved evaluation (I5) and tuning (I6) to the reference cohort, to test the benchmark's own model ranking. Full results are in Supplementary S2.

LASSO on the reference training set selected 12 variables at λ.1se (λ = 0.02667, against 0.02687 in the benchmark), seven of which overlapped with the benchmark's 12 (age, anion gap, RDW, chloride, PT, APS-III, CCI); at λ.min the overlap was 11 of 12.

Under nested cross-validation, six configurations were equivalent to the best model, with AUROC 0.786–0.791 (Table 3). The benchmark's two RF configurations and its SVM were significantly worse, with confidence intervals for the difference that excluded zero. Among the benchmark-parameter models, only XGBoost matched the best model.

Table 3: Reference cohort: nested cross-validation on the training set (1,568 patients; 5 × 10 folds) and the benchmark's reported validation AUCs.

Model	Benchmark validation AUC (Huang et al., 2025)	Nested-CV AUROC (mean ± SD)	AUPRC	Brier	Calibration slope	ΔAUROC vs best (95% CI)
XGBoost (benchmark parameters)	0.732	**0.791 ± 0.035**	0.559	0.150	0.96	reference
XGBoost (tuned)	—	0.790 ± 0.035	0.558	0.150	0.96	0.000 (−0.005, 0.006)
LR (L2, tuned)	—	0.788 ± 0.036	0.536	0.154	1.19	0.002 (−0.009, 0.014)
RF (tuned)	—	0.788 ± 0.036	0.548	0.154	1.48	0.002 (−0.006, 0.011)
SVM (tuned)	—	0.788 ± 0.036	0.541	0.153	0.90	0.003 (−0.008, 0.014)
LR (unpenalised)	0.773	0.786 ± 0.035	0.550	0.151	0.85	0.005 (−0.007, 0.016)
RF (benchmark, literal parameters)	0.780	0.767 ± 0.038	0.519	0.156	0.94	0.024 (0.006, 0.041)
RF (benchmark, R defaults)	0.780	0.724 ± 0.039	0.440	0.171	0.15	0.066 (0.043, 0.090)
SVM (benchmark parameters)	0.679	0.723 ± 0.044	0.497	0.163	0.96	0.068 (0.042, 0.093)

AUPRC baseline = 0.252. The first six rows are equivalent to the reference under the pre-specified rule, which selected the unpenalised LR (for this cohort the rule was not blind; Section 2.5). On the secondary held-out set (392 patients), the unpenalised LR had AUROC 0.803 (0.753–0.853) and calibration slope 0.96.

Three further results bear on the benchmark's pipeline.

- **RF overfitting (I6).** With R defaults (nine fully grown trees), the RF reproduced the benchmark's training AUC of 0.999 but had a validation AUC of 0.756 on a single split, a nested-CV AUROC of 0.724 and a calibration slope of 0.15, indicating severely overconfident predictions. With the literal parameters the training AUC was only 0.887. The benchmark's 0.999 is therefore consistent with fully grown trees, although other causes cannot be ruled out.
- **Variable selection outside cross-validation (I5).** Treating the LASSO selection as fixed, as the benchmark did, overstated the cross-validated AUROC of the unpenalised LR by 0.011 (0.793 vs 0.781), and of the nine configurations by 0.008–0.030 (single repeat of 10-fold cross-validation).
- **Preprocessing leakage (I5).** Fitting imputation and standardisation on all data, as in the benchmark, did not change the LASSO selection and changed model metrics by at most 0.001. Missingness was at most 7.04%, which probably explains why this form of leakage had negligible impact here.

A single validation set could not rank the models: on the 392-patient validation set, no pair of tuned models differed significantly after Holm correction (smallest adjusted p = 0.13).

3.4 Improved landmark model (main result)

This section reports the main estimate of the improved framework (I1–I6 combined) for the landmark cohort.

Nested cross-validation. LASSO selected 13–23 of the 56 candidates per outer fold (median 17). Age, anion gap, CCI, OASIS, APS-III, respiratory rate, RDW, maximum chloride and obesity were selected in all 50 folds; `wbc_diff_missing` was selected in 13 of 50. Of the six comorbidities, only obesity was ever selected. Six configurations were again equivalent (AUROC 0.794–0.801; Table 4), and the benchmark's RF and SVM configurations were again not.

Table 4: Landmark cohort: nested cross-validation on the training set (2,045 patients; 5 × 10 folds).

Model	AUROC (mean ± SD)	AUPRC	Brier	Calibration slope	ΔAUROC vs best (95% CI)
LR (L2, tuned)	**0.801 ± 0.026**	0.594	0.151	1.11	reference
LR (unpenalised)	0.799 ± 0.026	0.592	0.151	0.90	0.003 (−0.001, 0.006)
SVM (tuned)	0.798 ± 0.028	0.584	0.153	0.92	0.004 (−0.002, 0.009)
RF (tuned)	0.797 ± 0.028	0.591	0.155	1.46	0.004 (−0.005, 0.014)
XGBoost (tuned)	0.795 ± 0.028	0.596	0.152	0.99	0.006 (−0.004, 0.015)
XGBoost (benchmark parameters)	0.794 ± 0.029	0.597	0.152	1.02	0.008 (−0.002, 0.017)
RF (benchmark, literal parameters)	0.781 ± 0.029	0.560	0.157	1.09	0.020 (0.008, 0.033)
SVM (benchmark parameters)	0.756 ± 0.035	0.550	0.161	0.98	0.046 (0.027, 0.065)
RF (benchmark, R defaults)	0.740 ± 0.037	0.488	0.170	0.18	0.061 (0.039, 0.083)

AUPRC baseline = 0.266.

Model selection. The L2-penalised LR had the highest mean AUROC and was the reference M*. Both LRs were in the equivalent set, and the rule chose the L2 LR on calibration (mean |ln slope| 0.102 vs 0.109). This was a near tie: the two LRs differed in AUROC by 0.003, their slopes deviated from 1 by similar amounts in opposite directions (1.11 and 0.90), and the unpenalised LR was closer to 1 in three of five repeats. No claim of better discrimination is made for the selected model. Inner tuning chose C = 0.01 in all 50 folds.

Final model. Refitted on the whole training set (544 events), LASSO selected 22 variables (λ.1se = 0.0168; 24.7 events per selected variable) and inner tuning chose C = 0.01. The frozen model and the tuned RF comparator were evaluated once on the held-out set (512 patients, 136 deaths; Table 5, Figure 3).

Table 5: Landmark cohort: confirmatory held-out evaluation of the frozen models.

Metric	LR (L2), selected	RF (tuned), comparator
AUROC (95% CI)	0.750 (0.702–0.797)	0.752 (0.704–0.799)

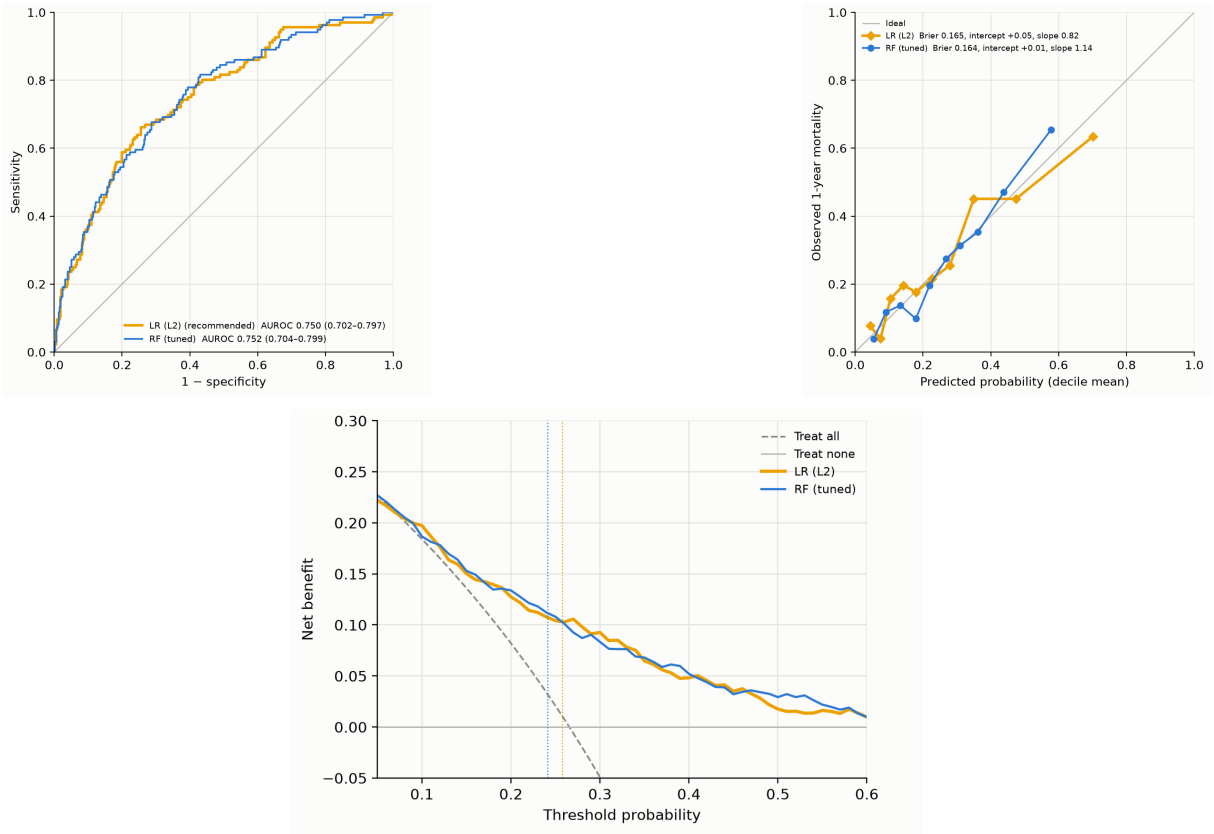


Figure 3: Frozen landmark models on the held-out set (512 patients, 136 deaths): (a) ROC curves, (b) calibration, (c) decision curves, with dotted vertical lines marking each model's frozen Youden threshold.

Metric	LR (L2), selected	RF (tuned), comparator
AUPRC (95% CI; baseline 0.266)	0.533 (0.459–0.611)	0.529 (0.460–0.615)
Brier / scaled Brier	0.165 / 0.153	0.164 / 0.161
Calibration intercept / slope	+0.049 / 0.82	+0.013 / 1.14
Threshold (OOF Youden)	0.258	0.241
Sensitivity / specificity	0.684 / 0.697	0.779 / 0.596
Precision / F1	0.449 / 0.542	0.411 / 0.538
Flagged as high risk	40.4%	50.4%

Paired DeLong test, LR − RF: −0.002 (−0.021, 0.016), p = 0.83.

Held-out estimate below the cross-validated estimate. The held-out AUROC was about 0.05 below the nested-CV estimate, and the nested-CV estimate lay above the upper limit of the held-out confidence interval. The RF comparator dropped in parallel (0.797 to 0.752). Post-hoc checks, which changed neither the frozen model nor the confirmatory result, did not point to overfitting: the frozen LR's apparent training AUROC was 0.817, only 0.016 above nested CV. When the OOF predictions were resampled 10,000 times to the held-out composition (136 deaths, 376 survivors), the mean AUROC was 0.800 (SD 0.021; 2.5th percentile 0.755), and 1.4% of resamples were at or below 0.750; this simulation ignores refitting variability and so understates the total variation. The largest standardised mean difference between training and held-out sets across the 22 selected variables was 0.11. The gap is therefore more likely sampling variation in a 512-patient set than overfitting, although other causes cannot be excluded. As pre-specified,

the model was not changed and the evaluation was not repeated; taken together, the landmark model's discrimination lies between 0.75 and 0.80.

3.5 Calibration and clinical utility (I7)

This section reports the metrics added under I7.

Calibration. Out of fold, the selected L2 LR had a calibration slope of 1.11 (predictions slightly too close to the mean) and the unpenalised LR 0.90. On the held-out set, the L2 LR's slope was 0.82 with intercept +0.049, so its predictions were slightly too extreme, the opposite direction to the cross-validated estimate. The tuned RF had out-of-fold slopes of 1.46 in the landmark cohort and 1.48 in the reference cohort (predictions shrunk towards the mean), and 1.14 on the held-out set. The benchmark's R-default RF had slopes of 0.18 and 0.15. Discrimination alone would not have revealed these differences, and the instability of the selected model's slope means that it must be recalibrated before use in a new population.

Overall accuracy. The held-out Brier score of the L2 LR was 0.165 (scaled Brier 0.153), similar to the RF comparator (0.164; 0.161).

Decision curves. On the held-out set, the L2 LR had net benefit above both treat-all and treat-none at 53 of 56 thresholds (0.08–0.60), and the tuned RF at 55 of 56 (0.06–0.60). In the reference cohort, the unpenalised LR was above both at 56 of 56 thresholds in nested cross-validation and on its held-out set.

Classification. At its out-of-fold Youden threshold (0.258), the landmark LR flagged 40.4% of held-out patients as high risk, of whom 44.9% died within one year (precision 0.449), with sensitivity 0.684 and specificity 0.697. A threshold of 0.5 would be inappropriate at this prevalence: in the reference cohort it flagged only 16.3% of patients, with sensitivity 0.44. The benchmark reported precision 0.735 and recall 0.858 for its RF; these values are internally consistent only if survival was the positive class, so they are not comparable with ours.

3.6 What each improvement changed (sensitivity analyses)

This section isolates the effect of the comorbidity-timing (I3) and white-cell differential (I2) modifications, and of the data release.

Table 6: Sensitivity analyses (nested cross-validation on the same outer splits; ΔAUROC = sensitivity − main, Nadeau–Bengio 95% CI).

Cohort	Analysis	Model	AUROC	ΔAUROC (95% CI)
Reference	Comorbidity timing	LR (unpenalised)	0.787	+0.001 (−0.006, 0.008)
Reference	Comorbidity timing	RF (tuned)	0.790	+0.002 (−0.006, 0.010)
Landmark	Prior-admission comorbidities only	LR (L2)	0.787	−0.014 (−0.023, −0.005)
Landmark	Prior-admission comorbidities only	RF (tuned)	0.789	−0.008 (−0.016, −0.0001)
Landmark	No white-cell differential	LR (L2)	0.802	+0.001 (−0.005, 0.007)
Landmark	No white-cell differential	RF (tuned)	0.795	−0.002 (−0.010, 0.005)

Comorbidity timing (I3). In the reference cohort, recomputing comorbidities from prior plus chronic

index-admission codes changed the CCI for 843 patients, and 304 of 611 myocardial-infarction diagnoses (about half) came only from acute index-admission codes. Discrimination was nevertheless unchanged, and CCI was still selected in every fold. In the landmark cohort, using prior admissions only lowered AUROC by 0.014 for the LR, with a confidence interval excluding zero. The largest change was in obesity, whose selection fell from 50 to 4 of 50 folds. This prior-only estimate (about 0.787) is the most conservative deployment scenario, because it does not assume that chronic index-admission codes were present at admission.

White-cell differential (I2). Removing all eight differential variables, including the missingness indicator, did not change discrimination, AUPRC, Brier score or calibration, and reduced the median number of selected variables from 17 to 13. The benefit of I2 is therefore in the cohort it allows, not in discrimination.

Data release. Excluding the 247 patients admitted in 2020–2022 from the reference cohort (new single 80/20 split) gave validation AUROCs of 0.799 for the unpenalised LR and 0.802 for the tuned RF, against 0.803 and 0.793 in the main split, and RDW and CCI remained the two most important predictors of the LR.

3.7 Predictors and interpretation

This section compares the predictors identified by the benchmark and by our models.

RDW and CCI ranked first and second in every model, and age ranked third in both LRs; the tuned RF ranked SAPS-II third (Table 7; Figure 4). In the landmark model (unpenalised refit, odds ratio per SD), higher risk was associated with RDW (OR 1.67, 95% CI 1.47–1.89), age (1.50, 1.30–1.75), CCI (1.44, 1.27–1.63), APS-III (1.38, 1.14–1.67), OASIS (1.23, 1.04–1.46), respiratory rate (1.20, 1.06–1.34) and minimum calcium (1.16, 1.02–1.32). Lower risk was associated with higher maximum chloride (0.82, 0.73–0.94) and coded obesity (yes vs no, 0.63, 0.49–0.81). Male sex (1.31, 1.03–1.66) and a missing white-cell differential (1.33, 1.04–1.71) were associated with higher risk. All variance inflation factors were below 5 (largest 4.74, SAPS-II), and SAPS-II, LODS, NLR and SII were not significant once the other scores and ratios were included.

Table 7: Top-5 predictors by mean |SHAP|.

Rank	Benchmark (RF) (Huang et al., 2025)	Reference: LR (unpenalised)	Reference: RF (tuned)	Landmark: LR (L2)
1	RDW	RDW	RDW	RDW
2	Neutrophils	CCI	CCI	CCI
3	CCI	Age	SAPS-II	Age
4	Chloride (max)	APS-III	APS-III	APS-III
5	PT (min)	OASIS	Chloride (max)	Obesity

3.8 Exploratory extensions

This section reports the extensions proposed in the project brief; all results come from the development set.

None of the extensions clearly improved on the static LR (Table 8). Stacking had the highest AUROC, 0.805 against 0.799, a difference of 0.005 (95% CI −0.001 to 0.011; p = 0.077, Holm-adjusted p = 1.00).

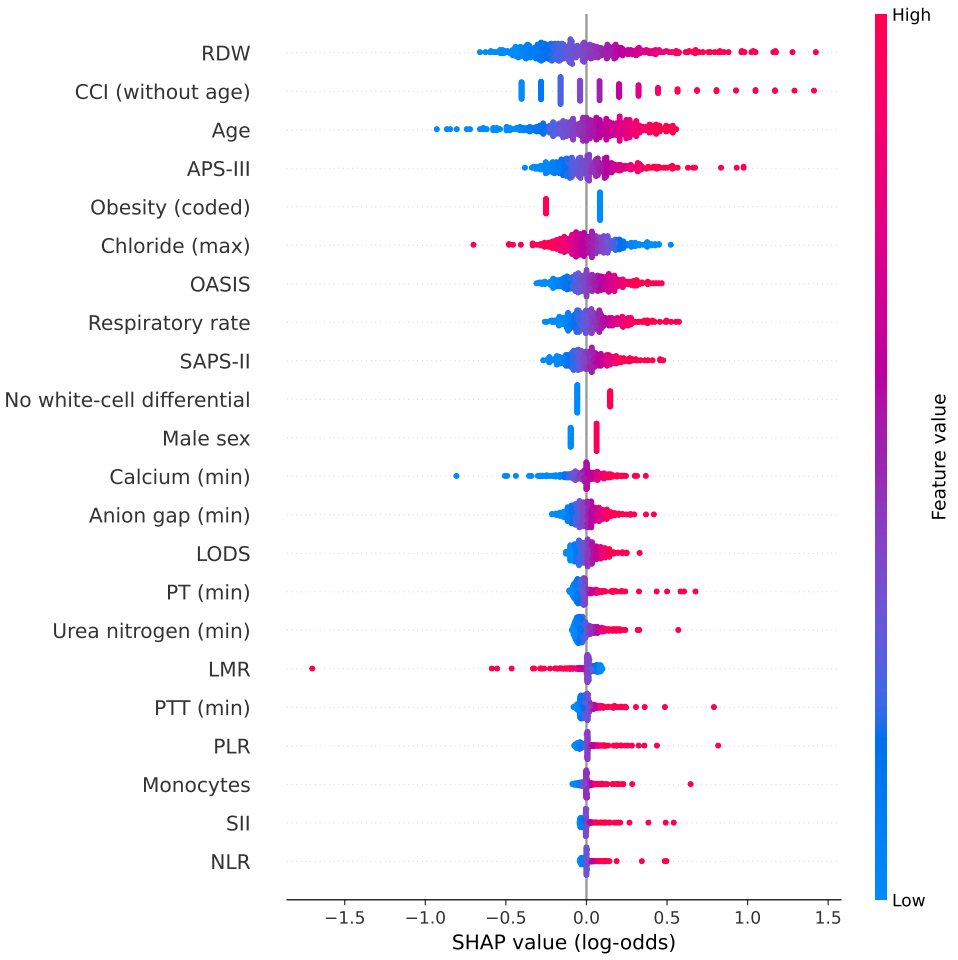


Figure 4: SHAP summary of the frozen landmark LR (L2) on the held-out set (512 patients), log-odds scale, predictors ordered by mean |SHAP|. Colour shows the predictor value. For patients without a white-cell differential, the differential-derived variables take their imputed training means and appear as a cluster near zero. The corresponding plots for the reference cohort are in Supplementary S2.

Its direction was consistent, with higher AUROC in 41 of 50 folds and a higher pooled out-of-fold AUROC in all five repeats, and its out-of-fold calibration slope (1.05) was closer to 1 than that of LR (0.90); its net benefit, however, was higher at only 31 of 56 thresholds. Trajectory features entered every fit (median 4.5 selected per fold, range 1–13; SpO2 variability in 48 of 50 folds) but lowered AUROC slightly for LR, RF and XGBoost (by 0.004, 0.001 and 0.002). Prior diagnoses were available for 1,183 development patients (57.8%). Neither the burden variables nor the embedding improved LR, and the embedding's small gain for RF (0.002 over the static RF) came with a worse Brier score and a calibration slope further from 1 (1.35). Adding all extensions to the stacking ensemble changed AUROC by less than 0.001. All comparisons had Holm-adjusted p = 1.00; the full results for 14 models are in Supplementary S12.

Table 8: Exploratory extensions on the landmark development set (fixed-parameter recipe, 5 × 10 folds; ΔAUROC against the matched reference on the same folds, with Nadeau–Bengio-corrected 95% CI).

Model	AUROC (mean ± SD)	AUPRC	Brier	Reference	ΔAUROC (95% CI)
Static LR (L2)	0.799 ± 0.026	0.607	0.151	—	—
Static stacking	0.805 ± 0.027	0.616	0.150	Static LR	0.005 (−0.001, 0.011)
LR + trajectories	0.796 ± 0.026	0.602	0.153	Static LR	−0.004 (−0.010, 0.003)
LR + diagnosis burden	0.799 ± 0.027	0.607	0.151	Static LR	0.000 (−0.002, 0.002)
LR + burden + embedding	0.796 ± 0.027	0.606	0.152	LR + burden	−0.003 (−0.011, 0.004)
Stacking, all extensions	0.805 ± 0.027	0.616	0.150	Static stacking	0.000 (−0.008, 0.008)

4 Discussion

4.1 Principal findings

Starting from the benchmark's framework, we corrected seven issues and evaluated the result without leakage. The improved landmark model predicts one-year mortality at a defined time, 24 h after ICU admission, from information available then, and keeps the third of patients that the benchmark's blood-count criterion excluded. Its discrimination was moderate, with a nested-CV AUROC of 0.801 and a confirmatory held-out AUROC of 0.750. Simple LR performed as well as RF, XGBoost and SVM once every family was tuned inside the folds. The benchmark's conclusion that RF is superior was not reproduced: its RF configuration was significantly worse than the best model, and its training AUC of 0.999 is reproduced by a forest of nine fully grown trees that memorises the training data. The leading predictors were stable across cohorts and models: RDW, comorbidity burden, age, illness severity and low chloride.

4.2 Impact of each improvement

Table 9 summarises what each modification changed. Some changed the conclusions; others changed little and are reported for completeness.

Table 9: Observed effect of each improvement.

Improvement	Observed effect
I1 Landmark at 24 h	Gives the model a defined time of use and removes a criterion that depends on the future. The landmark model's nested-CV AUROC (0.801) was similar to that of the reference cohort (0.786–0.791), although the cohorts differ and this is not a controlled comparison.
I2 Keep patients without a differential	Retained 873 patients (34.1%), whose mortality was 28.6% against 25.5% (p = 0.09); the missingness indicator had OR 1.33 (1.04–1.71) but was selected in only 13 of 50 folds. Removing all differential variables changed AUROC by +0.001 (-0.005, 0.007): no effect on discrimination.
I3 Comorbidity timing	Changed the CCI for 843 reference patients; about half of the myocardial-infarction diagnoses (304 of 611) rested only on acute codes. AUROC did not change (+0.001). Relying on prior admissions only lowered AUROC by 0.014 and removed obesity from the model (50 to 4 of 50 folds), so the effect is mainly on interpretation.
I4 Definitions	A fixed 365-day window gave 25.2% mortality, against 42.8% when any recorded death is counted. The full hypertension definition removed a spurious protective effect (OR 0.49 vs 1.07) without changing the LASSO selection. With absolute counts, neutrophils were not selected at λ.1se.
I5 Nested evaluation	Selecting variables outside cross-validation overstated AUROC by 0.008–0.030 in a single-repeat analysis (0.011 for the unpenalised LR). Preprocessing leakage changed metrics by at most 0.001: no material effect at this missingness. A single validation set could not separate the tuned models, and the held-out estimate was 0.05 below the cross-validated one, showing the variability of single-set estimates.
I6 Tuning all families	The largest effect. The benchmark's RF configurations reached 0.724 and 0.767, significantly below the best model (0.791), whereas the tuned RF (0.788) was equivalent to it. With tuning, the apparent advantage of RF disappeared.
I7 Calibration and decision curves	Revealed differences that discrimination hides: the R-default RF had slope 0.15, and the selected LR's slope moved from 1.11 out of fold to 0.82 on the held-out set, so recalibration is required. Net benefit exceeded both default strategies at 53 of 56 thresholds.

4.3 Comparison with the benchmark

RF did not outperform the other families because, once all were tuned inside the folds, the differences between them were within 0.005 in the reference cohort and 0.008 in the landmark cohort. The benchmark's reported advantage of RF rested on a single 263-patient validation set, on which a confidence interval spans about $\pm$0.05 (0.723–0.837 for its RF), and on a training AUC that reflects memorisation rather than performance.

The landmark model's held-out AUROC of 0.750 is below the benchmark's 0.780, but the two numbers are not comparable, for four reasons. First, the cohorts differ: the landmark cohort includes patients without a white-cell differential, does not exclude patients with short hospital stays, and comes from a later data release. Second, the outcomes probably differ: the benchmark's abstract names in-hospital death, yet its mortality of 38.2% is far above in-hospital mortality in our cohort and is consistent with counting deaths without a fixed window. Third, our estimate is confirmatory: the model was frozen before a single held-out evaluation, whereas the benchmark's validation set also served for model selection. Fourth, the held-out confidence interval (0.702–0.797) includes 0.780, and the same pipeline gave 0.786–0.791 in the reference cohort and 0.801 under nested cross-validation in the landmark cohort. The appropriate conclusion is that discrimination is moderate, about 0.75–0.80, in both studies.

4.4 Biological plausibility of the main predictors

RDW was the strongest predictor in every analysis, in agreement with the benchmark and with studies linking RDW to mortality in critically ill patients (Bazick et al., 2011). Elevated RDW reflects heterogeneous erythropoiesis driven by inflammation, oxidative stress, nutritional deficiency and renal dysfunction, which are prevalent in long-standing diabetes. **CCI** captures the cardiovascular and renal burden that typically accompanies DN (Feldman et al., 2019), and its importance persisted after recomputing it from chronic codes only. **Low chloride**, associated with higher risk in both our work and the benchmark, is a recognised marker of poor outcome in critical illness (Tani et al., 2012). The apparently protective effect of **coded obesity** was concentrated in patients whose obesity code appeared only during the index admission: their one-year mortality was 18.8% (n = 416), against 26.4% (n = 443) when obesity was coded in a prior admission and 28.6% (n = 1,698) without an obesity code. This suggests a documentation artefact rather than a true "obesity paradox", and it should be tested with measured body-mass index.

4.5 Limitations

- **Single-centre, retrospective data and no external validation.** The landmark held-out set has only 136 events, which limits the precision of the confirmatory estimate.
- **DN identified from ICD codes.** Their accuracy is unknown, and codes carry no timestamps. The main landmark estimate assumes that chronic index-admission codes were present at admission; using prior admissions only lowers AUROC to about 0.787.
- **Unstable calibration.** The slope was 1.11 out of fold and 0.82 on the held-out set, so the model must be recalibrated before use in a new population.
- **Only first-day features.** Treatments were not used, within-day dynamics were tested only in the exploratory extensions, and cause of death is unavailable.
- **Exploratory extensions.** They were run after the held-out results had been seen, with fixed rather than tuned hyperparameters, and were not evaluated on held-out data; a small gain from stacking cannot be excluded.
- **Non-blind model selection in the reference cohort.** The selection rule was written after exploratory analyses of the reference cohort, and its held-out set had been used in those analyses; only the landmark analyses were fully pre-specified.

4.6 Future work

- **Temporal validation.** Train on earlier and test on later MIMIC-IV year groups, and validate externally in data with one-year follow-up.
- **Recalibration and thresholds.** Recalibrate the model in the target population and choose thresholds for specific clinical uses.
- **Objective measurements.** Use measured body-mass index to test the obesity finding.
- **Richer predictors and models.** Add HbA1c and treatment variables, and test stacking with tuned base learners on new data. Clinical notes, SHAP interaction values and counterfactual explanations, also proposed for this project, were not run.

5 Conclusion

We took a published machine-learning framework for one-year mortality in ICU patients with diabetic neuropathy and corrected its cohort definition, predictor definitions and evaluation. The resulting landmark model uses only information available 24 h after ICU admission and was frozen before a single confirmatory evaluation. Under leakage-free nested cross-validation, LR matched tree ensembles, and the benchmark's reported RF advantage reflected an overfitted configuration and a single small validation set. RDW, comorbidity burden and age were consistently the strongest predictors. Exploratory trajectory features, diagnosis embeddings and stacking added little or nothing to discrimination. The landmark model's discrimination is moderate (AUROC 0.75–0.80), and it should be recalibrated locally before clinical use.

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6 Supplementary material

- **S1. Four cohort definitions.** Four readings of "first ICU admission at first hospitalization", with counts and one-year mortality.
- **S2. Reference cohort: complete results.** Nested cross-validation of all nine configurations, LASSO selection frequencies, the model-selection steps, the secondary held-out estimate, thresholds, decision curves, coefficients and SHAP.
- **S3. Full baseline tables** of the reference and landmark cohorts.
- **S4. Comparison with the benchmark's Table 1** and the mixed-unit diagnostic for the white-cell differential.
- **S5. Hypertension definitions** and their associations with one-year death.
- **S6. LASSO cross-validation curve and coefficient path.**
- **S7. Exploratory single-split results** (all models, paired DeLong comparisons, thresholds, calibration, LASSO-only nesting).
- **S8. Comorbidity-timing recode**: acute codes removed and changes in each comorbidity.
- **S9. Implementation details.** Global random seed 42 and repeat seeds; LASSO implementation (λ grid of 100 values from $\lambda_{\max} = \max_j |\mathbf{x}_j^\top (\mathbf{y} - \bar{y})|/n$ down to $10^{-4}\lambda_{\max}$; liblinear intercept feature set to 1,000 so that the intercept penalty is negligible; binomial deviance as the criterion, whereas the benchmark's text names mean squared error but its figure shows binomial deviance); the benchmark's hyperparameters and their mapping to scikit-learn and xgboost; all tuning grids.
- **S10. Benchmark-style versus leakage-free preprocessing**: LASSO selection, validation AUROC and SHAP.
- **S11. Code availability and reproducibility**: pipeline scripts, seeds and the frozen-model specification. All code, aggregate results and this manuscript's sources are available at https://github.com/hongfangyu123/BMI5101-DN-Mortality-P16; patient-level MIMIC-IV data are not included and require credentialed PhysioNet access.
- **S12. Exploratory extensions: complete results.** All 14 models (LR, RF, XGBoost and stacking with static, trajectory, burden and embedding inputs), all 45 paired comparisons, pooled out-of-fold calibration, decision-curve comparisons and trajectory-feature selection frequencies.